Tableau

Setup

Download Tableau public

Download the software:

https://www.tableau.com/zh-tw/downloads/public/mac

https://www.tableau.com/zh-tw/downloads/public/pc64

Sample data:

https://learning-platform-danger.s3.ap-east-1.amazonaws.com/BI+Visualization/conduct.csv

https://learning-platform-danger.s3.ap-east-1.amazonaws.com/BI+Visualization/exam.csv

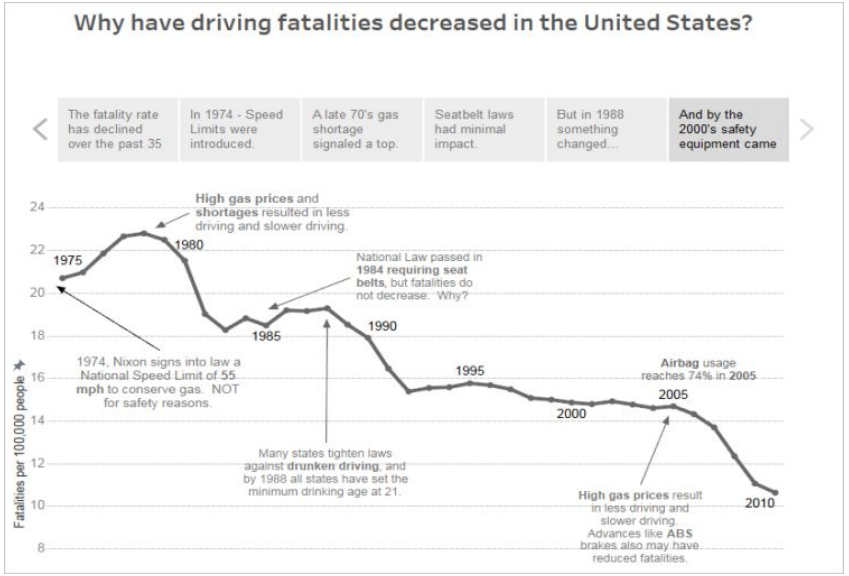
https://learning-platform-danger.s3.ap-east-1.amazonaws.com/BI+Visualization/student.csv

https://learning-platform-danger.s3.ap-east-1.amazonaws.com/BI+Visualization/demerit.csv

Basic usage

Key objects

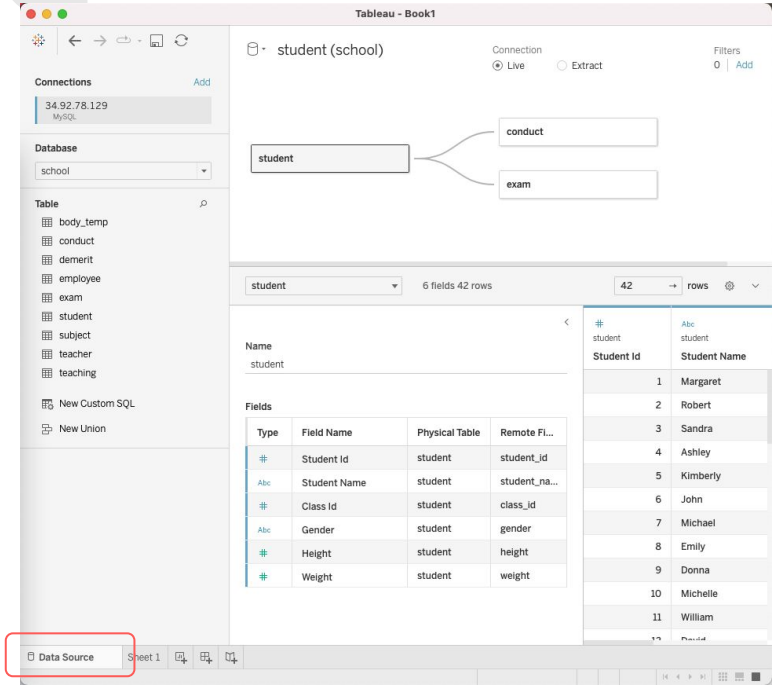
1. Book
2. Data source
3. Sheet
4. Dashboard
5. Story



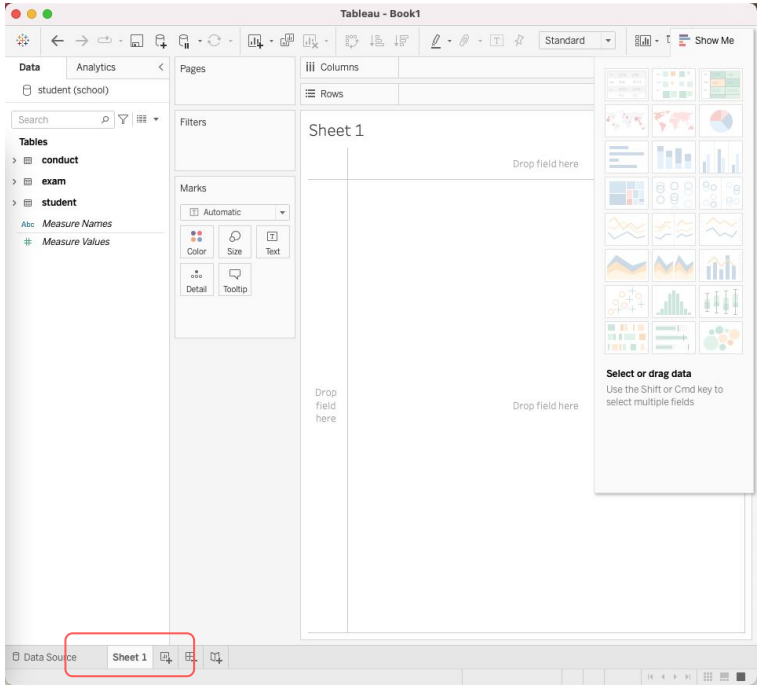
A story is a sequence of visualizations

Workspace

The whole thing is a “book”, which can be saved in a file



Manage data sources



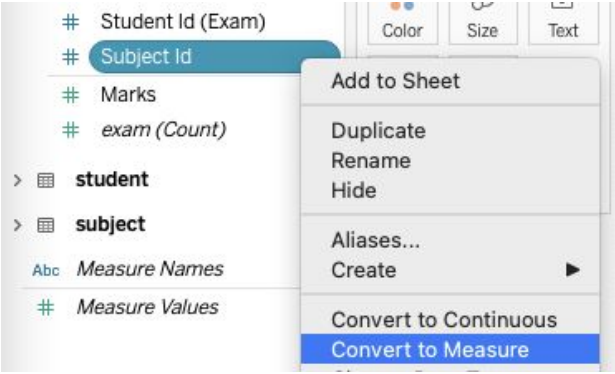
Work on worksheets

exam_id	exam_description	subject_id	student_id	marks
1	First Term	1	1	67
1	First Term	1	2	71
1	First Term	1	3	65
1	First Term	1	4	83
1	First Term	1	5	62
1	First Term	1	6	78

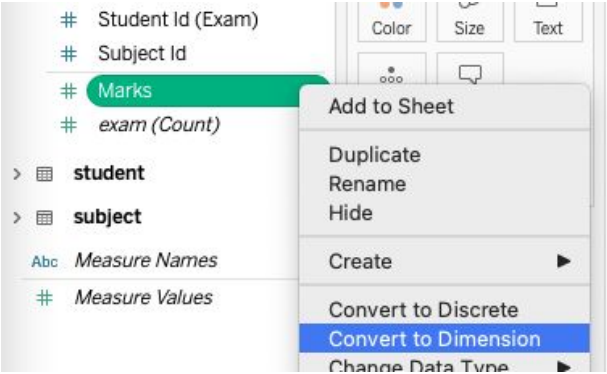
Dimension vs Measure

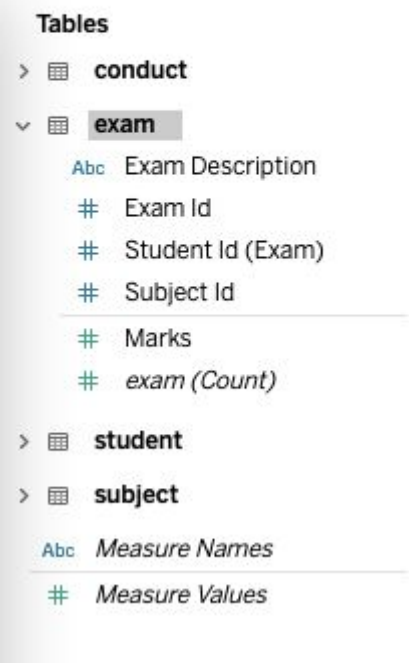
Tableau automatically detect if columns are dimension or measure, you can change the setting of the detection is wrong.

Upper part



Lower part





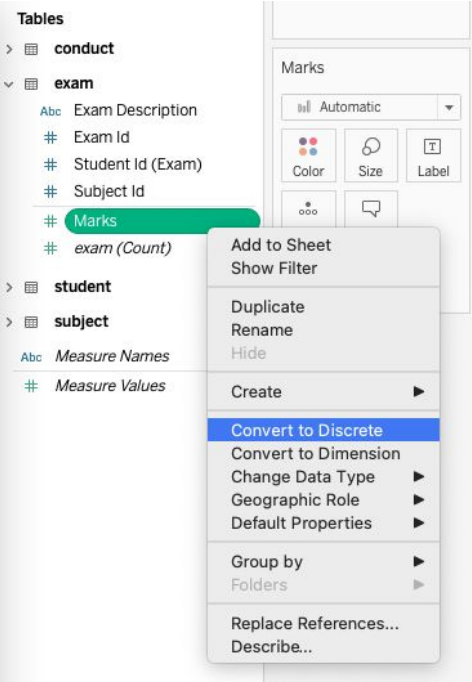
Discrete or continuous

Examples of discrete variables:

- Group numbers
- Categorical variables

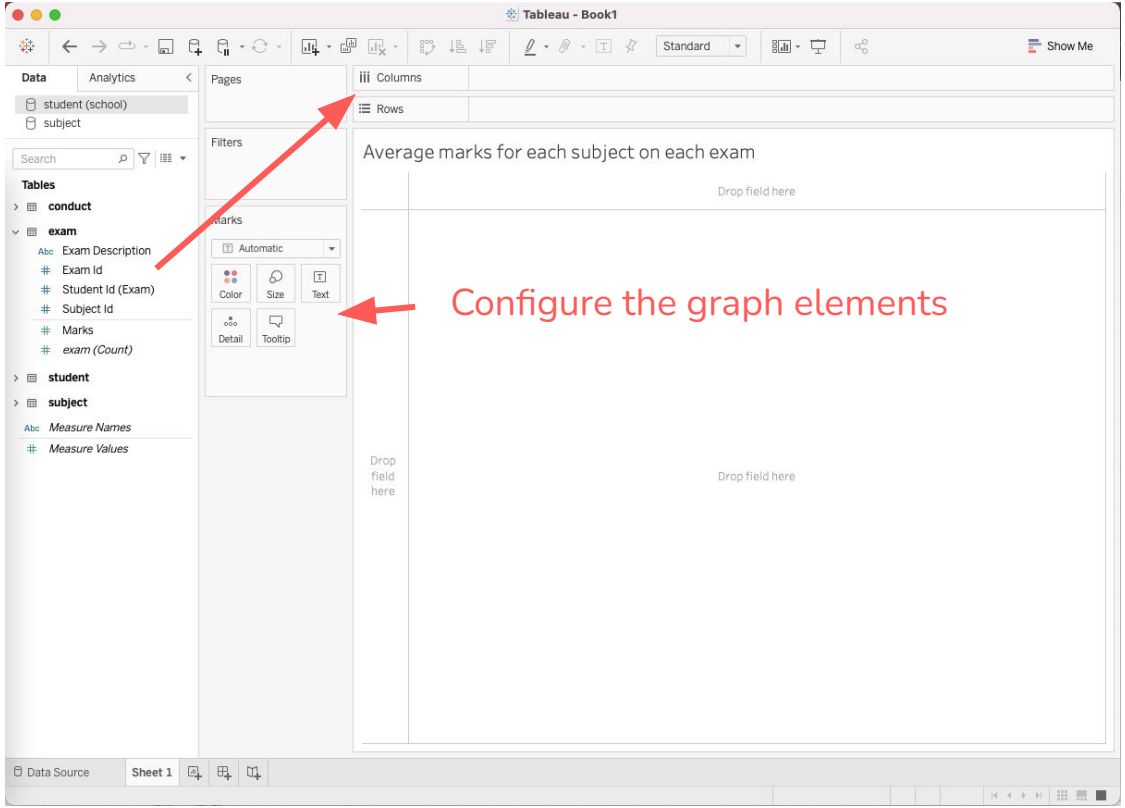
Examples of continuous variables:

- Height/Weight
- Percentage



How to plot?

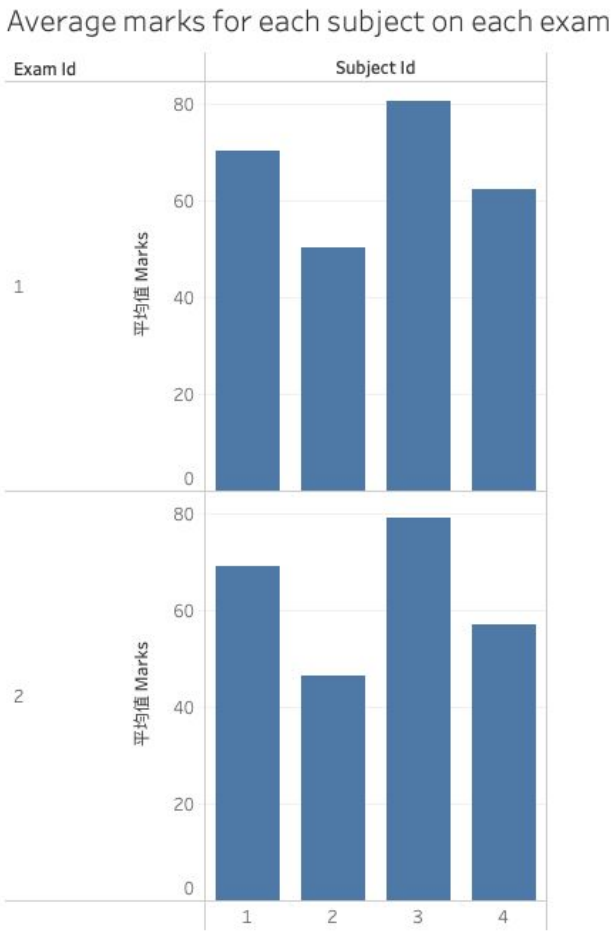
Drag dimensions or measures to columns or rows



Configure the graph elements

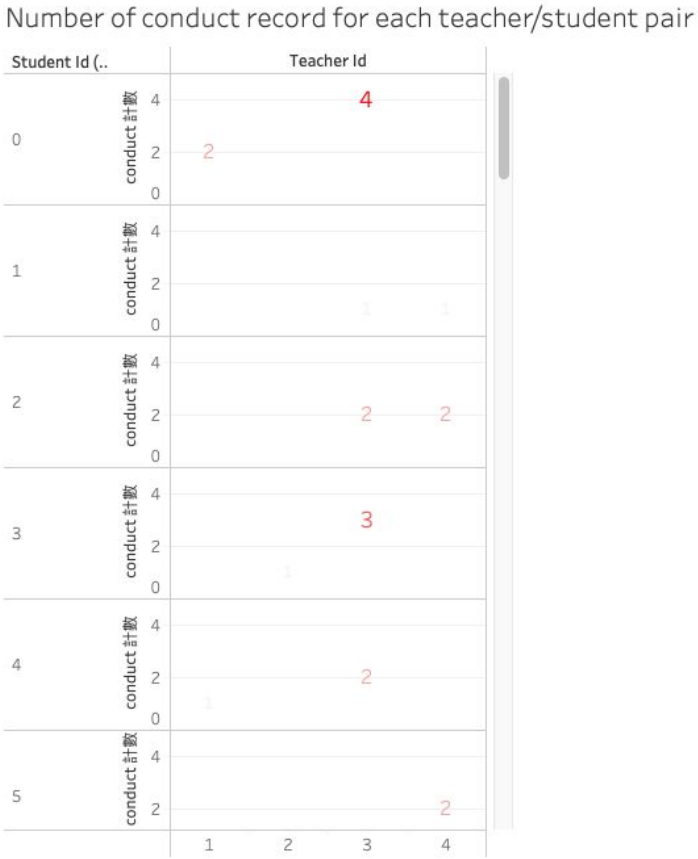
Exercise

Plot this graph ->



Exercise

Plot this graph ->

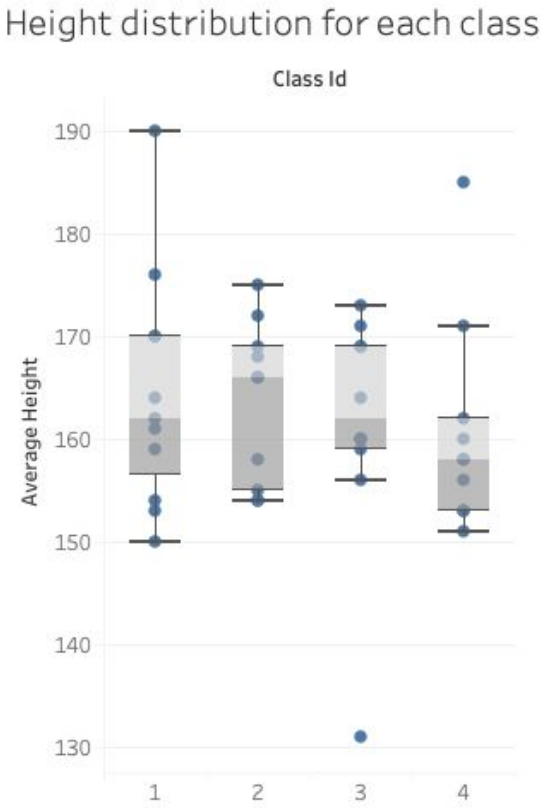


Exercise

How about we give each student their name?

Exercise

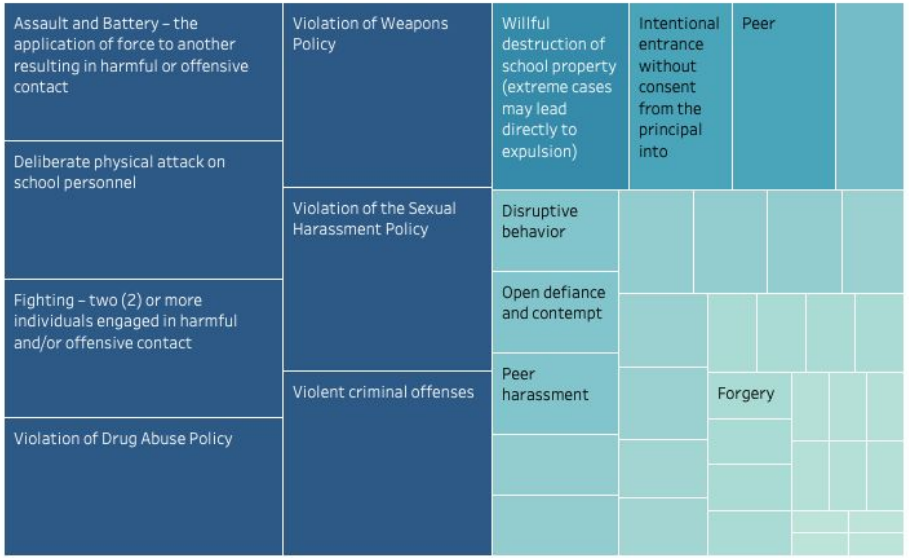
Plot a box diagram to indicate the height distribution of each class



Exercise

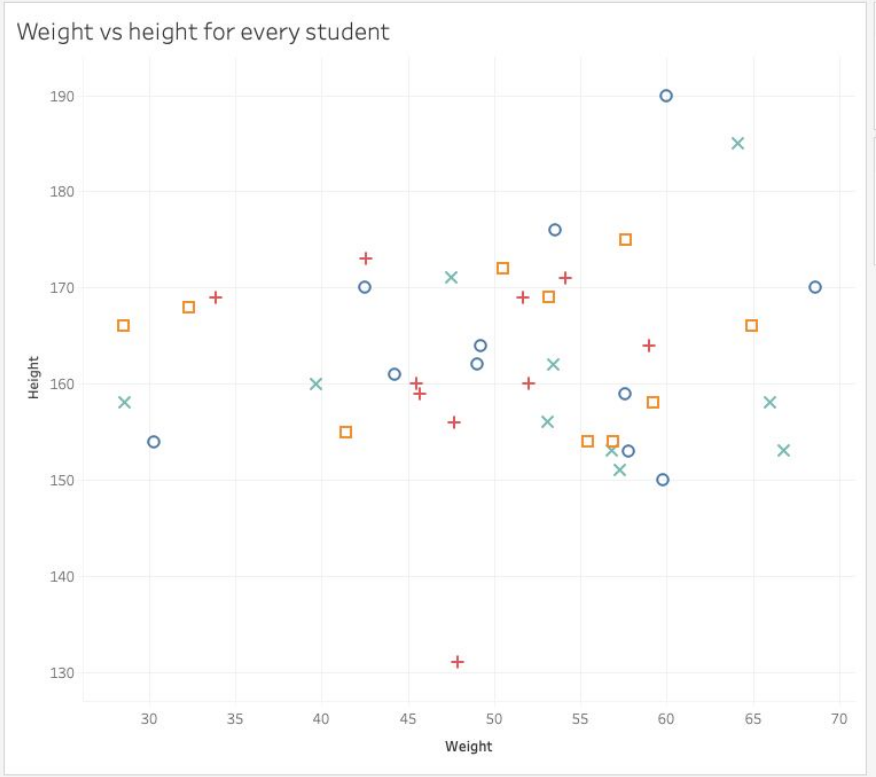
To show the severity of each kind of misconduct.

Severity of misconducts



Exercise

How about scatter plot?
Different marker for
different class.



Exercise

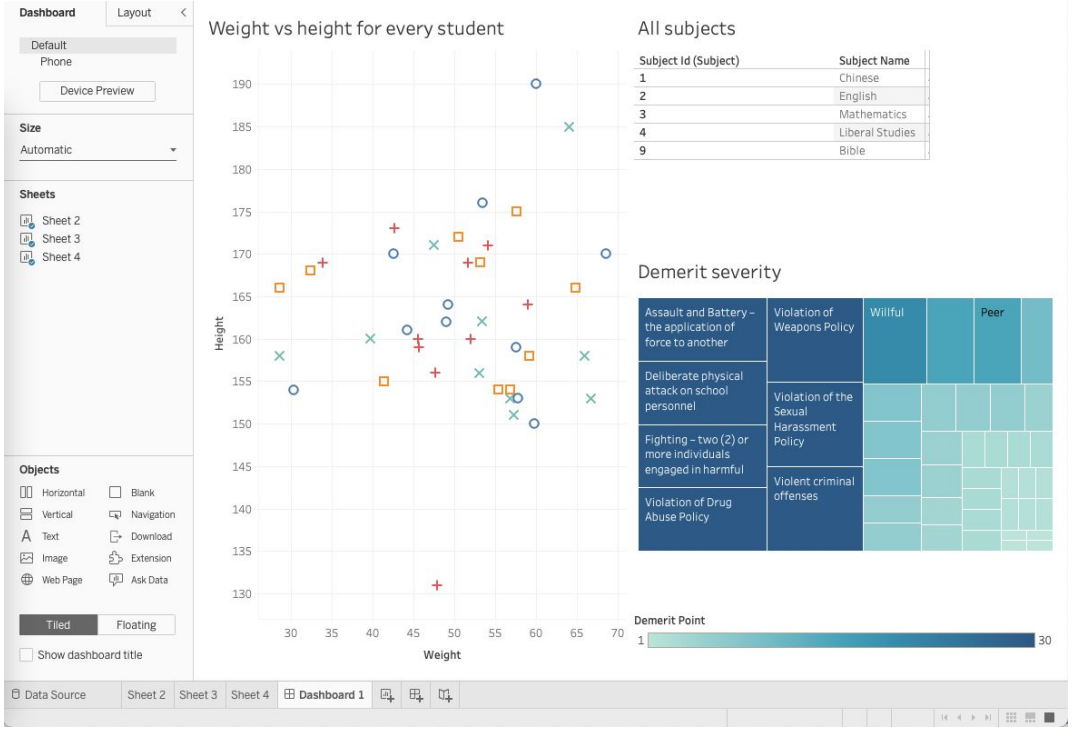
How about a simple text table?

All subjects

Subject Id (Subject)	Subject Name
1	Chinese
2	English
3	Mathematics
4	Liberal Studies
9	Bible

Dashboard

A dashboard group worksheets into one page.



Share your work

Share the book file

The workbook file is a self contained file.

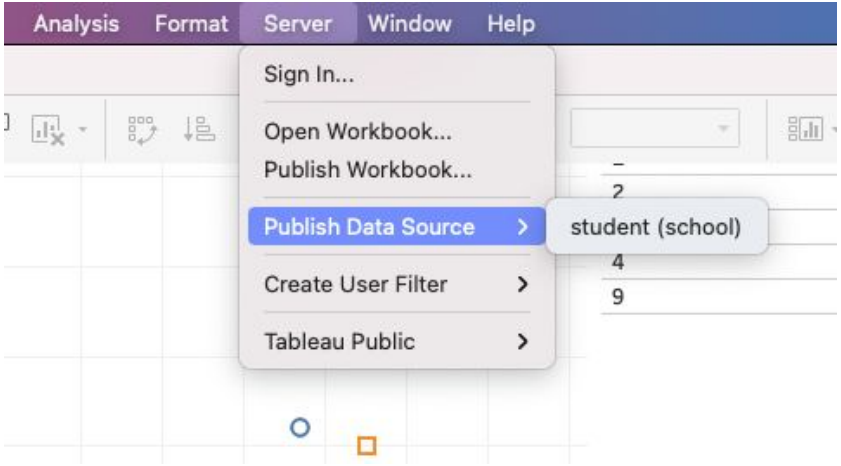
It can be opened by other people in other computers.

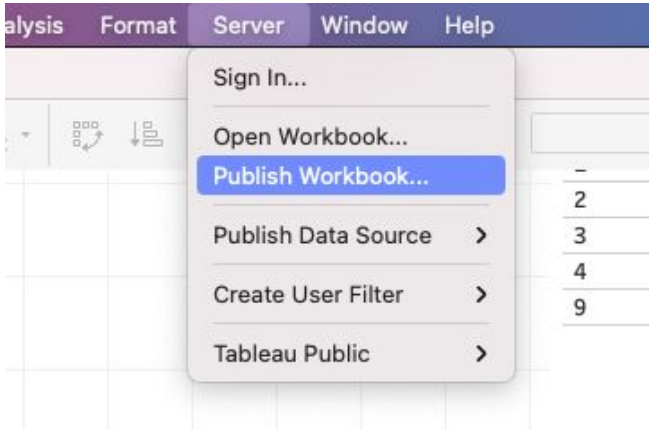
Share online

Tableau comes with an online service - Tableau online (not free).

Steps:

1. Publish data source
2. Publish workbook

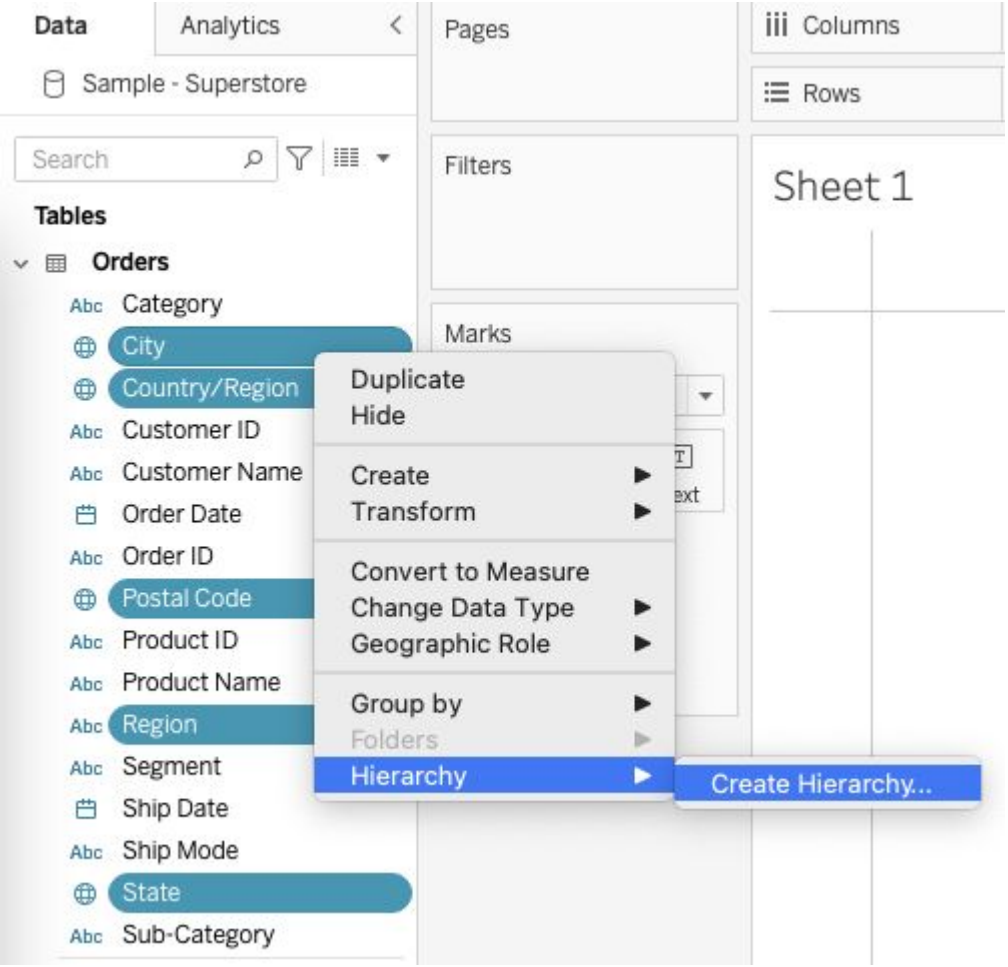




Advanced usage

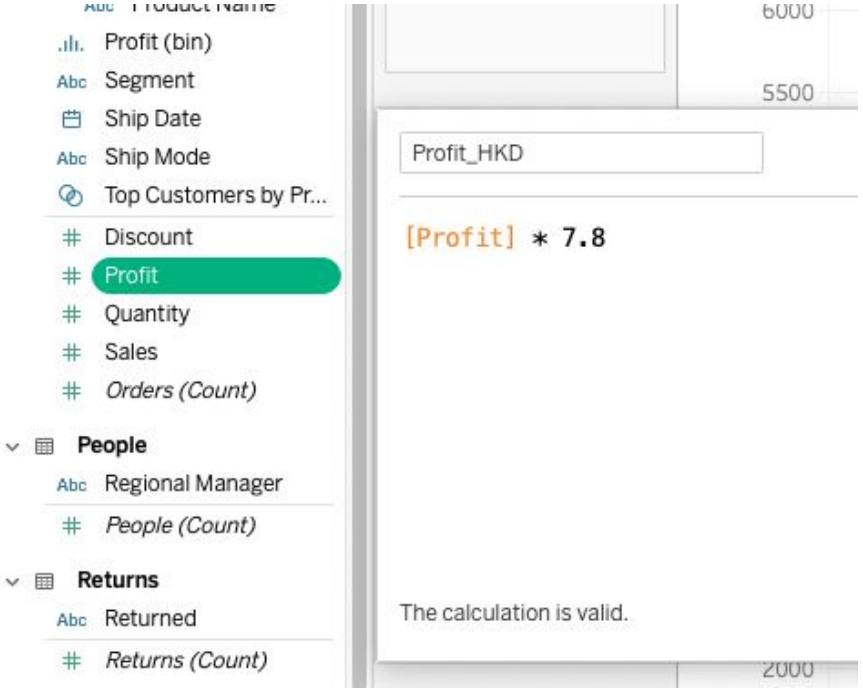
Hierarchy

We can group fields into a hierarchy for better management.

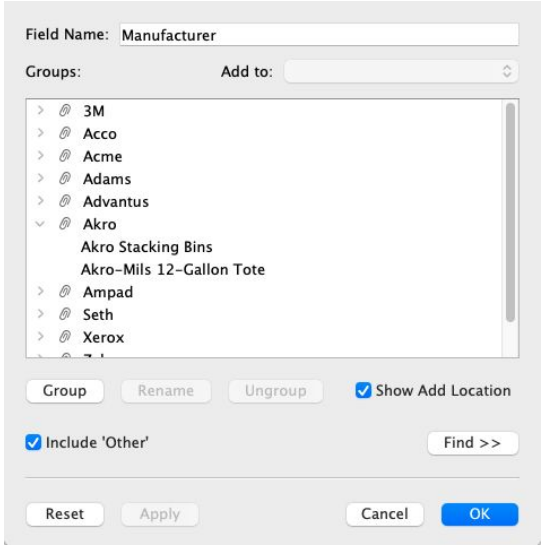


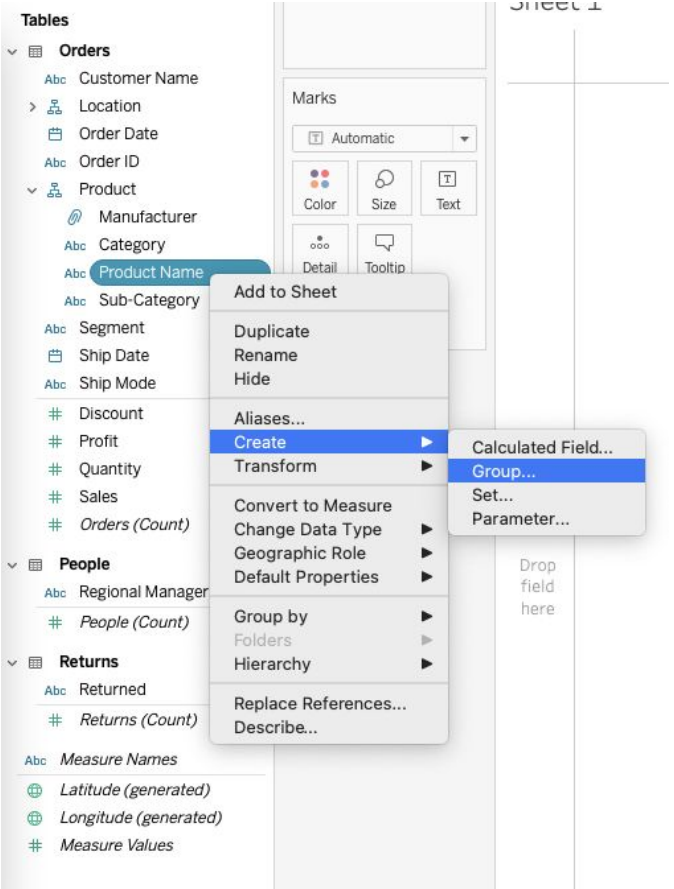
Calculated Field

We can create new field with a formula. The formula can contain one or more existing field denoted by `[fieldname]`.



Group





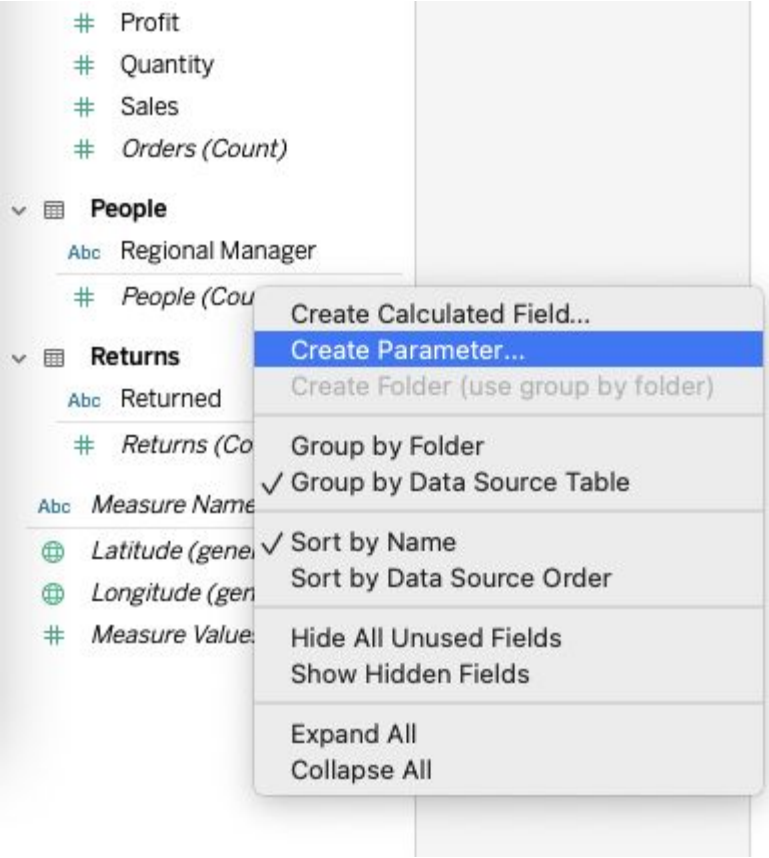
Very similar to Calculated Field, but instead of creating new values with a function, “Group” creates a mapping between original value to a “Group”.

E.g. “Akro Stacking Bins” -> “Akro”

Parameters

Like “Variables”.

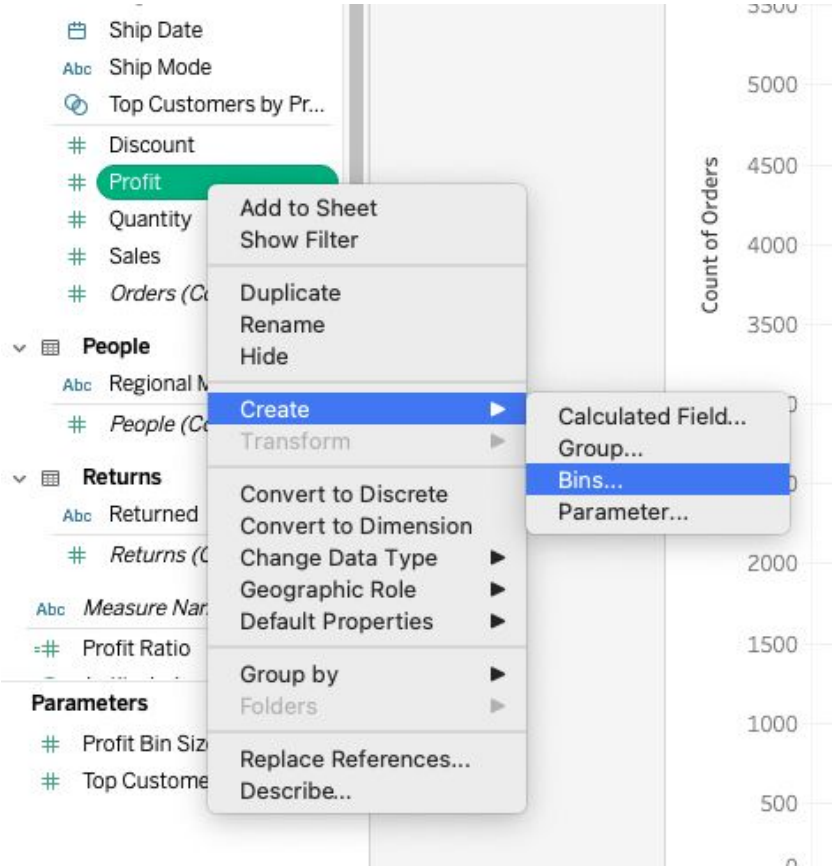
It is a variable that can be used in calculation and thus affect the graph layout.



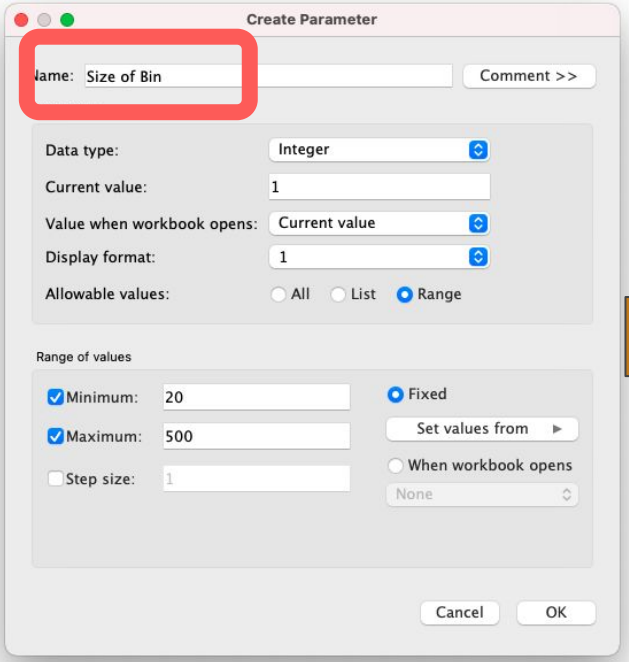
Bins

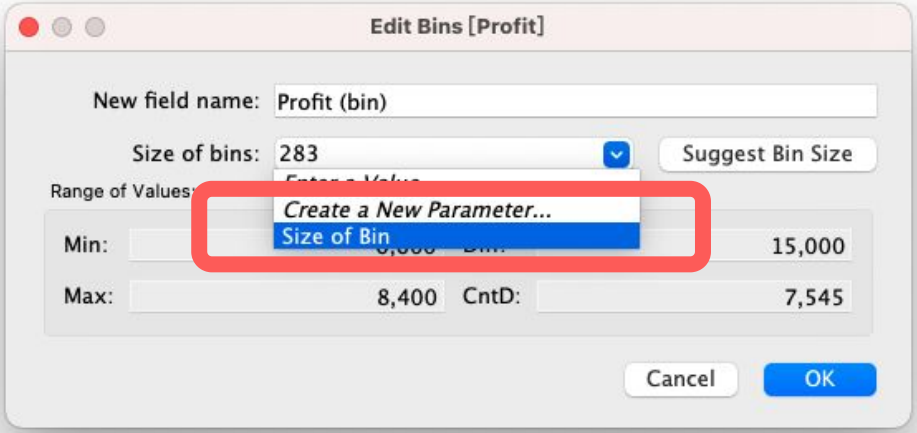
Another kind of calculated field. It turns a column of continuous values into discrete “bins”.

The biggest use case is to create histograms.

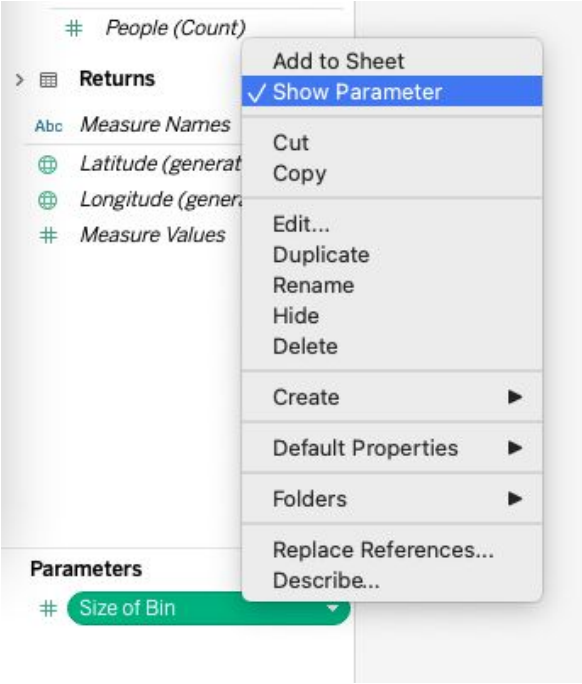


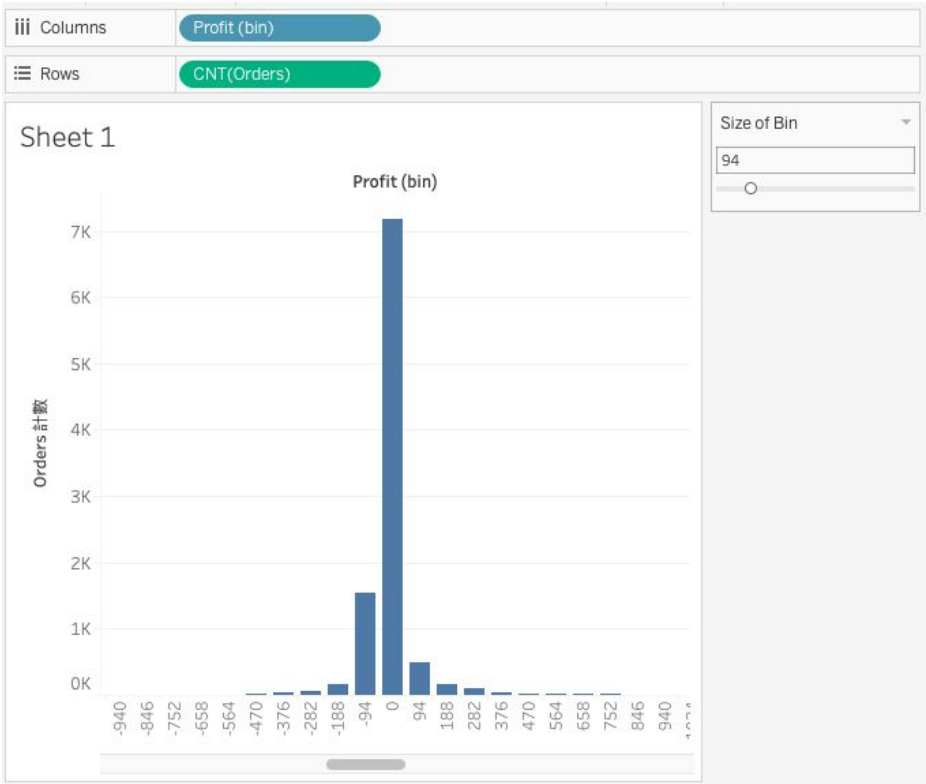
Example: Using parameter in Bins setting





Example: Create a dynamic graph with parameter



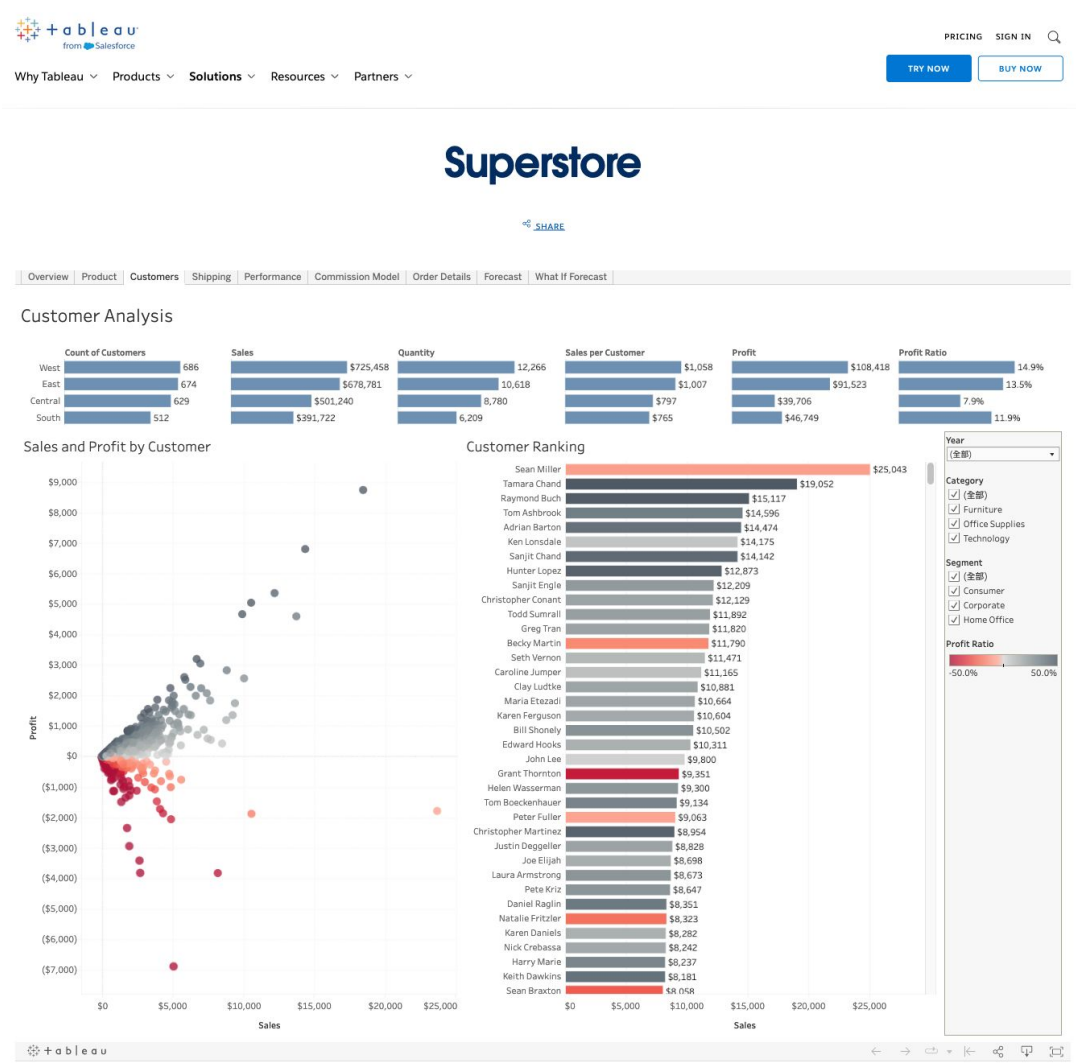


Project

Case study

https://www.tableau.com/solutions/gallery/superstore

In this case study we will try to replicate the Customer Analysis dashboard.

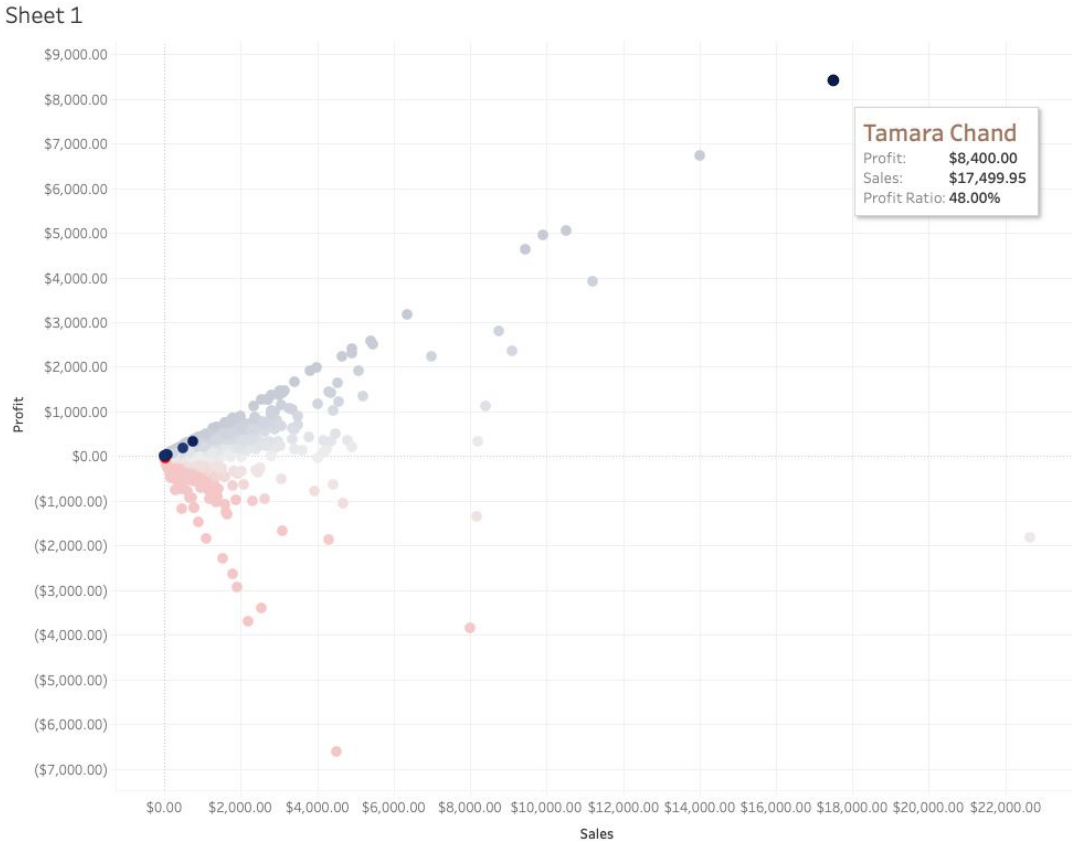


Dashboard interaction

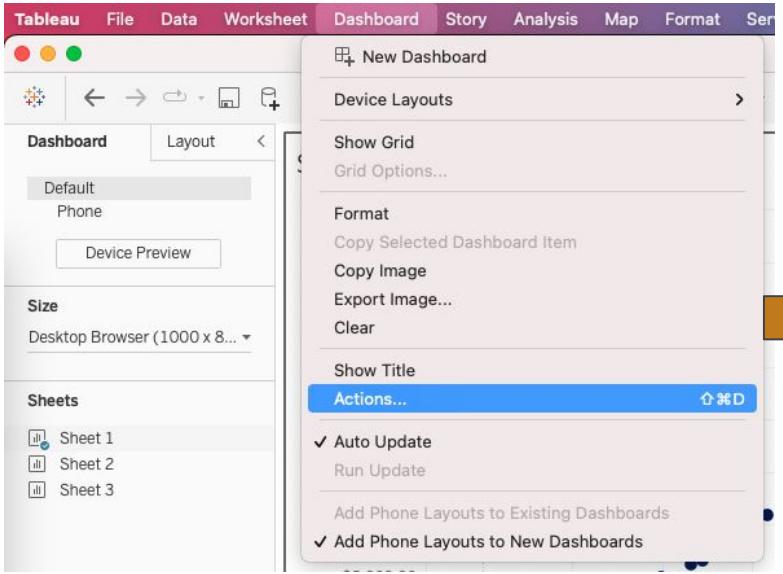
Users can retrieve extra information by performing dashboard interaction, e.g. Hover/Click.

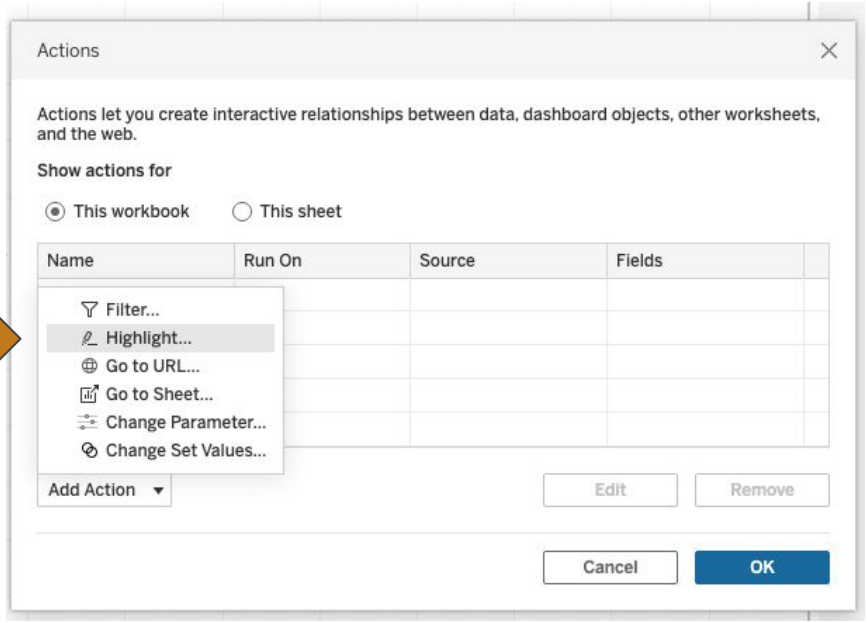
Example goal:

To select the points with the same customer Id.

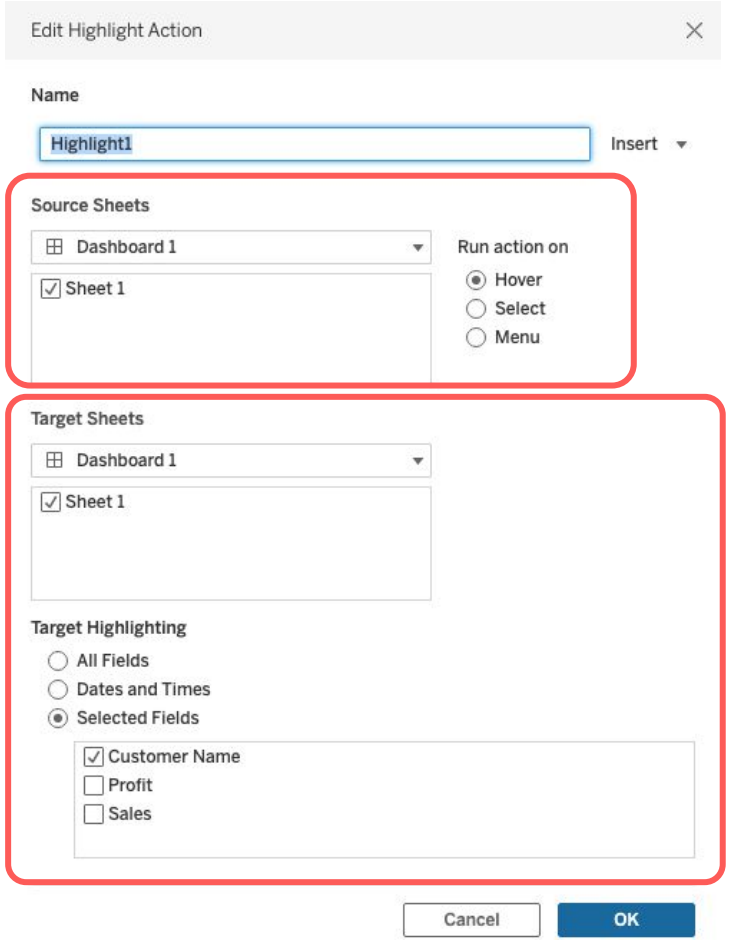


Dashboard interaction





Dashboard interaction

Trigger by:

Perform action on: